Solve with us 1.1 - Not Graded

This assignment will not be graded and is only for practice.

Key Points:

- Vectors in $\mathbb{R}^2$ are represented by ordered pairs (a, b).
- The first element in the ordered pair denotes the X-coordinate and the second one denotes the Y-coordinate in the Cartesian plane.
- The addition of two vectors $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2) \in \mathbb{R}^2$, is given by $v_1 + v_2 = (x_1 + x_2, y_1 + y_2)$.
- The scalar multiplication of a vector $v = (x, y) \in \mathbb{R}^2$ with a scalar $c \in \mathbb{R}$, is given by $cv = (cx, cy)$.

1 point

Let $v_1 = (2, -4)$ and $v_2 = (-1, 2)$ be two vectors in $\mathbb{R}^2$. Which of the following options are true?

[Hint: Recall that vector addition and scalar multiplication are done coordinatewise.]

- [ ] $2v_2 = (-2, 4)$
- [ ] $\frac{1}{2}v_1 = (1, -2)$
- [ ] $v_1 + 2v_2 = (0, 0)$
- [ ] $2v_1 + v_2 = (0, 0)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$2v_2 = (-2, 4)$

$\frac{1}{2}v_1 = (1, -2)$

$v_1 + 2v_2 = (0, 0)$

1 point

Choose the set of correct options using Figure M2W1SWU1.

[Hint: Recall that vector addition and scalar multiplication are done coordinate-wise.]

- [ ] $2A$ is the vector $(2, 4)$.
- [ ] $3B$ is the vector $(6, 9)$.
- [ ] $A + B$ is the vector $(3, 5)$.
- [ ] $A - B$ is the vector $(-1, -1)$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$2A$ is the vector $(2, 4)$.

$3B$ is the vector $(6, 9)$.

$A + B$ is the vector $(3, 5)$.

$A - B$ is the vector $(-1, -1, -1)$.

2. Key points:

- The addition of two vectors $V_1 = (a_1, b_1, C_1)$ and $V_2 = (a_2, b_2, C_2)$ are done coordinate-wise, as follows:
- $$V_1 + V_2 = (a_1, b_1, C_1) + (a_2, b_2, C_2) = (a_1 + a_2, b_1 + b_2, C_1 + C_2)$$
- The scalar multiplication of a vector $V = (a, b, c)$ by a scalar α is given by:
$$\alpha(a, b, c) = (\alpha a, \alpha b, \alpha c)$$

1 point

Let $V_1 = (1, 0, 0)$, $V_2 = (0, 1, 0)$ and $V_3 = (0, 0, 1)$ be three vectors and $a, b,$ and c be three real numbers (scalars). Then, which of the following is (are) true?

- [ ] $(a, b, c) = aV_1 + bV_2 + cV_3$
- [ ] $(a, b, c) = abV_1 + bcV_2 + caV_3$
- [ ] $(a, 0, c) = aV_1 + cV_2 + 0V_3$
- [ ] $(a, 0, c) = aV_1 + 0V_2 + cV_3$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(a, b, c) = aV_1 + bV_2 + cV_3$

$(a, 0, c) = aV_1 + 0V_2 + cV_3$

3. Key Points:

- Addition of matrices $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ and

$B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$ is given by

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \\ a_{31} + b_{31} & a_{32} + b_{32} & a_{33} + b_{33} \end{bmatrix}$$

a22 + b22a32 + b32a13 + b13a23 + b23a33 + b33]

- If $R = [a\ b\ c]$ and $C = \begin{bmatrix} d \\ e \\ f \end{bmatrix}$, then the product RC is given by

  $$[ad + be + cf]$$

- Suppose $A = \begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$ and $B = [C_1\ C_2\ C_3]$, where R_i's denote the rows of matrix A and C_i's denote the columns of matrix B. Moreover, assume that the number of columns of A and the number of rows of B are the same. The product AB is given by,

  $$AB = \begin{bmatrix} R_1C_1 & R_1C_2 & R_1C_3 \\ R_2C_1 & R_2C_2 & R_2C_3 \end{bmatrix}$$

- Addition of two matrices A and B is defined if both A and B have the same number of rows and the same number of columns. If both the matrices A and B have m rows and n columns, then the matrix $A + B$ also has m rows and n columns

- If A is an $m \times n$ matrix and B is an $n \times p$ matrix, then AB is well-defined and is an $m \times p$ matrix.

1 point

Suppose $P = \begin{bmatrix} 3 & -1 & 7 \\ 4 & 0 & 1 \\ 2 & -5 & 2 \end{bmatrix}$, $Q = [1\ 4\ {-9}]$, $R = [0\ {-3}\ 10]$, $D = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}$

[Hint: If A is a matrix of order $m \times n$ and B is a matrix of order $n \times p$, then the order of AB is $m \times p$.]

☐ The matrix PD is of order 3×1.

☐ The matrix PD is of order 1×3.

☐ The matrix QD is of order 3×3.

☐ The matrix QD is of order 1×1.

☐ The matrix DQ is of order 3×3.

☐ The matrix DQ is of order 1×1.

☐ The product QD is not defined.

☐ The product QR is not defined.

☐

The addition $P + Q$ is not defined.

☐

The addition $P + D$ is not defined.

No, the answer is incorrect.

Score: 0

Feedback:

- P is a 3 × 3 matrix, D is a 3 × 1 matrix. Think about the order of P D.
- Q is a 1 × 3 matrix, D is a 3 × 1 matrix. Think about the order of QD and DQ.
- Q is a 1 × 3 matrix, R is a 1 × 3 matrix. Think whether is it possible to define QR or not.
- Figure out which pair of matrices has the same number of rows as well as the same number of columns.

Accepted Answers:

The matrix PD is of order 3×1.

The matrix QD is of order 1×1.

The matrix DQ is of order 3×3.

The product QR is not defined.

The addition $P + Q$ is not defined.

The addition $P + D$ is not defined.

1 point

Suppose $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 5 \\ 3 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 2 & 3 \\ 2 & 5 & 4 \end{bmatrix}$. Which of the following options are correct?

[Hint: Multiplication of matrices $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ and $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$ is given by

$$AB = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} & a_{11}b_{13} + a_{12}b_{23} + a_{13}b_{33} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} & a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} \\ a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31} & a_{31}b_{12} + a_{32}b_{22} + a_{33}b_{32} & a_{31}b_{13} + a_{32}b_{23} + a_{33}b_{33} \end{bmatrix}$$

Similarly, BA can be calculated.]

☐

$AB = \begin{bmatrix} 5 & 10 & 11 \\ 10 & 26 & 29 \\ 11 & 29 & 34 \end{bmatrix}$

☐

$AB = \begin{bmatrix} 5 & 10 & 11 \\ 10 & 29 & 26 \\ 11 & 26 & 34 \end{bmatrix}$AB = [5101110292611263 4]

☐

$BA = \begin{bmatrix} 10 & 9 & 14 \\ 9 & 13 & 22 \\ 14 & 22 & 45 \end{bmatrix}$BA = [1091491322142245]

☐

$BA = \begin{bmatrix} 10 & 9 & 14 \\ 9 & 22 & 13 \\ 14 & 13 & 45 \end{bmatrix}$BA = [1091492213141345]

No, the answer is incorrect.
Score: 0
Feedback:

Observe that B = A^T (where A^T denotes the transpose of A).

Accepted Answers:

$AB = \begin{bmatrix} 5 & 10 & 11 \\ 10 & 29 & 26 \\ 11 & 26 & 34 \end{bmatrix}$AB = [51011102926112634]

$BA = \begin{bmatrix} 10 & 9 & 14 \\ 9 & 13 & 22 \\ 14 & 22 & 45 \end{bmatrix}$BA = [1091491322142245]

1 point

Suppose $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$A = [100−1] and $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$B = [0010] Which of the following options are true?

[Hint: Calculate A^2 A2 and B^2 B2]

☐
$A^2 = I$A2 = I

☐
$A^2 = A$A2 = A

☐
$B^2 = I$B2 = I

☐
$B^2 = 0$B2 = 0

No, the answer is incorrect.
Score: 0
Feedback:

Correctly solving this question will give a hint to solve Week 4 Lecture 1.2 : Activity Question 6.

Accepted Answers:

$A^2 = I$A2 = I
$B^2 = 0$B2 = 0

4. Key points:

- Consider a system of linear equations as follows:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}$$

Let the matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$,

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \text{ and } b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

- A is called Coefficient matrix.

- Suppose there are m number of equations and n number of variables in the system of linear equations, then the coefficient matrix will be an $m \times n$ matrix, x will be an $n \times 1$ and b will be an $m \times 1$ matrix.

1 point

Consider a system of linear equations (System 1):

$$\begin{aligned} -2x_1 + 3x_2 + x_3 &= 1 \\ -x_1 + x_3 &= 0 \\ 2x_2 &= 5 \end{aligned}$$

If the matrix representation of system (1) is $Ax = b$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, then

[Hint: If a variable is absent in an equation then the coefficient of the variable must be taken as 0.]

☐ $A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$

☐ $b = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$

☐ $A = \begin{bmatrix} -2 & -1 & 0 \\ 3 & 0 & 2 \\ 1 & 1 & 0 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$

☐

$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$

$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$

1 point
Consider a system of equations:

$$\begin{aligned} 2x_1 + 3x_2 &= 6 \\ -2x_1 + kx_2 &= d \\ 4x_1 + 6x_2 &= 12 \end{aligned}$$

Choose the set of correct options.

[Hint: Observe that third equation is a multiple of the first one (Dividing by 2 from both the sides of the third equation gives the first equation). So it is enough to check the solutions for the first and second equation.]

- [ ] $Ax = b$ represents the above system, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$, and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$
- [ ] The system has no solution if $k = -3$, $d = 0$.
- [ ] The system has a unique solution if $k = 3$, $d = 0$.
- [ ] The system has infinitely many solutions if $k = -3$, $d = 6$.
- [ ] The system has infinitely many solutions if $k = -3$, $d = -6$.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$Ax = b$ represents the above system, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$,

and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$

The system has no solution if $k = -3$, $d = 0$.
The system has a unique solution if $k = 3$, $d = 0$.
The system has infinitely many solutions if $k = -3$, $d = -6$.

5. Key points:

- Let A be a 2×2 matrix as follows:

  $$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

  $$det(A) = ad - bc$$

- Let A be a 3×3 matrix as follows:

  $$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$det(A) = a_{11} \times det\left(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}\right) - a_{12} \times det\left(\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}\right) + a_{13} \times det\left(\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}\right)$$

i.e., $det(A) = a_{11} \times (a_{22}a_{33} - a_{32}a_{23}) - a_{12} \times (a_{21}a_{33} - a_{23}a_{31}) + a_{13} \times (a_{21}a_{32} - a_{22}a_{31})$

(Expanding with respect to first row)

Find the determinant of the matrix $\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$

[Hint: Use the formula for finding the determinant of a matrix of order 2×2.]

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 1

1 point

1 point

Let $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 2 & 1 & 1 \end{bmatrix}$ be a 3×3 matrix. Which of the following is(are) correct? [Hint: Determinant can be calculated by expanding with respect to different rows or columns.]

- [ ] $det(A) = 3 \times det\left(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}\right) - 2 \times det\left(\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}\right) + 2 \times det\left(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}\right)$

]) + 2 × det ([2231])

☐
$det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$det(A) = 3 × det ([3121]) + 2 × det ([2122

]) + 2 × det ([2231])

☐
$det(A) = -2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}) + 3 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix})$det(A) = −2 × det ([2121]) + 3 × det ([3221

]) − 2 × det ([3221])

☐
$det(A) = 2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}) - 3 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix})$det(A) = 2 × det ([2121]) − 3 × det ([3221

]) + 2 × det ([3221])

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$det(A) = 3 × det ([3121]) − 2 × det ([2221

]) + 2 × det ([2231])

$det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$det(A) = 3 × det ([3121]) + 2 × det ([2122

]) + 2 × det ([2231])

$det(A) = -2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}) + 3 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix})$det(A) = −2 × det ([2121]) + 3 × det ([3221

]) − 2 × det ([3221])

6. Key points:

- Row operations and relation with determinants:

  - Type 1: Interchanging two rows of a matrix changes the sign of the determinant.
  - Type 2: Multiplying a real number with a row and adding it to some other row does not change the determinant.
  - Type 3: The determinant of a new square matrix obtained by multiplying a real number cc with a row of square matrix of order nn, is cc times the determinant of the original matrix.

- Same types of column operations, as discussed above for row operations, gives same type of relation with the determinants.
- If two rows or two columns of a matrix are equal, then the determinant of the matrix is 00.
- If all the elements of any row or any column of a matrix are 00, then the determinant of the matrix is 0 0.
- $det(A) = det(A^T)$det(A) = det(AT), where A^T AT denotes the transpose of AA.
- $det(AB) = det(BA) = det(A)det(B)$det(AB) = det(BA) = det(A)det(B).

1 point
Let AA be a $3 \times 3$3 × 3 matrix with non-zero determinant. If $det(2A) = k\ det(A)$det(2A) = k det(A), then what will be the value of kk?
[Hint: If a scalar (c)(c) is multiplied with one row of a matrix AA, then the determinant of the new matrix will be cc times the determinant of AA.]

○ 22
○ 44
○ 88
○ 1616

No, the answer is incorrect.
Score: 0
Accepted Answers:
88

1 point

Let AA be a $2 \times 2$2 × 2 matrix, which is given as $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$[a11a21a12a22]. Define the following matrices :

$$B = \begin{bmatrix} a_{11} - a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}$$ B = [a11 − a21a21a12 − a22a22], $$C = \begin{bmatrix} a_{11} - a_{12} & a_{12} \\ a_{21} - a_{22} & a_{22} \end{bmatrix}$$ C = [a11 − a12a21 − a22a12a22],

$$D = \begin{bmatrix} a_{11} + a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}$$ D = [a11 + a21a21a12 − a22a22], $$E = \begin{bmatrix} a_{11} - a_{21} & a_{12} + a_{22} \\ a_{21} & a_{22} \end{bmatrix}$$ E = [a11 − a21a21a12 + a22a22]

Which of the matrices among $B, C, D,$ and EB, C, D, and E have the same determinant as that of the matrix AA, for any real numbers $a_{11}, a_{12}, a_{21}, a_{22}$ a11, a12, a21, a22?
[Hint: Try to observe whether $B, C, D,$B, C, D, and EE can be obtained from AA by row or column operations of Type $2$2 mentioned in the key points.]

○ BB and DD
○ BB and EE
○ BB and CC
○ DD and EE
○ CC and EE

No, the answer is incorrect.
Score: 0
Feedback:
• B can be obtained by a row operation of Type 2.
• C can be obtained by column operation of Type 2.
• D cannot be obtained by a row operation or column operation of Type 2. Check the determinant of D and see whether it matches with determinant of A or not.
• E cannot be obtained by a row operation or column operation of Type 2. Check the determinant of E and see whether it matches with determinant of A or not.

Accepted Answers:
BB and CC

1 point

If all the elements of a $3 \times 3$3 × 3 real matrix AA are the same, then which of the following is (are) correct?

[Hint:

- $\because$ If two rows or two columns of a matrix are equal, then the determinant of the matrix is 0.
- $\because$ $det(A) = det(A^T)$, where A^T denotes the transpose of A].

- [ ] Determinant of matrix A is 0.
- [ ] Determinant of matrix A cannot be determined from the given information.
- [ ] Determinant of matrix A will be the sum of the elements of a row.
- [ ] Determinant of matrix $A + A^T$ is 0, where A^T denotes the transpose of A.
- [ ] Determinant of matrix $A + A^T$ cannot be determined from the given information, where A^T denotes the transpose of A.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Determinant of matrix A is 0.

Determinant of matrix $A + A^T$ is 0, where A^T denotes the transpose of A.

1 point

Let $A, B,$ and C be 3×3 real matrices. Which of the following options is (are) correct?

[Hint: $\because$ $det(A) = det(A^T)$, where A^T denotes the transpose of A.

$\because$ $det(AB) = det(BA)$.]

- [ ] $det(ABC) = det(A)det(B)det(C)$
- [ ] $det(A^3) = det(A)^3$
- [ ] $det(A + B + C) = det(A) + det(B) + det(C)$